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**MODELING OF A HYBRID ENERGY SYSTEM INTEGRATED WITH AN
INTERMITTENT POWER GRID IN DEVELOPING COUNTRIES**

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ABSTRACT

Electricity in most developing countries is inefficient, erratic and unreliable. Recent research work on grid-connected hybrid system in developing countries have not considered frequent electricity outage hours per month experienced by consumers. Effort is made to address this critical factor and HOMER software was used to model an intermittent power grid integrated with a hybrid energy system. This shortcoming is addressed by assuming generator component in HOMER to run as grid. This is made possible after some modification to fuel content, minimum load capacity ratio and scheduling capacity of the generator. A city in Nigeria was used to demonstrate this approach, but it can be replicated for any location in developing countries where there is frequent electricity outage. Outcome of the model shows that integrating an intermittent power grid with a hybrid energy system offers a competitive cost of energy, reduction in noise pollution and improved use of renewable energy for electricity generation in developing countries.

INTRODUCTION

There exist vast underutilized renewable energy resources in the world. Developing countries have many renewable energy resources which are most times never used or under-utilized. In most developing countries it is difficult to satisfy power

demand all year long by hydro sources and fossil fuel- coal, gas, diesel etc. It is therefore necessary to interconnect other renewable/alternative energy sources [1]. Renewable energy resources are defined as energy whose future availability is not affected by their use, although some resources can cease if harvesting is not performed in a sustainable manner. Renewable energy technologies transform a renewable energy resource into heat, cooling, electricity or mechanical energy [2]. Sources of renewable energy are Solar (radiation from Sun), Wind, Water, Geothermal, Timber, Fruits and vegetables etc. This article focuses on wind and solar resources for generation of electricity combine with conventional means of generating electricity- diesel generator and grid. This type of system is known as Hybrid Energy system.

According to World Bank, countries that lack strong amounts of industrialization, infrastructure, and technology can be considered as developing countries. They are mainly characterized with low economic standard of living. Energy (Electricity) markets in developing countries often do not function efficiently. Poor organizational structures, lack of sound institutional and legal frameworks, and huge electrical grid losses impair the provision of efficient, reliable and affordable energy. Energy supply limitations prevent people from meeting their basic needs, and thus slow their economic growth [3].

Grid-connected Hybrid Energy System (HES) - for developing countries- offers a solution to electricity problems. HES is necessary to meet the needs of billions of people who still lack access to basic, modern energy services while simultaneously participating in a global transition to clean, low-carbon energy systems. Access to reliable, affordable and socially acceptable energy services is a prerequisite to alleviating extreme poverty and meeting other societal development goals [4].

Emissions from developing countries are growing rapidly and are contributing to environmental problems, such as climate change and poor air quality that put the health and prosperity of people around the world at grave risk [4].

Other reasons for HES include: Rising consumption and a transition from traditional sources of energy to commercial forms of energy, steady improvement in the power and efficiency of energy technologies; and a tendency toward fuel diversification and de-carbonization, especially for electricity production; Improved pollution control and lower emissions[4].

HOMER software from National renewable Energy Laboratory was used to analyze the model. HOMER simulates and optimizes a system by making energy balance calculations for every hour of each of the 8760 hours in a year. It seeks to determine the least cost combination of components that meet the load demand. Decision variables for this model are Photovoltaic generator, wind turbine, Diesel generator and grid [5]. This software will help to analyze an optimal system that will meet daily load demand by combining intermittent grid with renewable energy.

NOMENCLATURE

HES	Hybrid Energy system
PV	Photovoltaic
STC	Standard Test conditions
STP	Standard Temperature and Pressure
P_{pv}	Power output of the PV array [kW]
Y_{pv}	Rated power output of the PV array under STC [kW]
f_{pv}	Derating factor [%]

G_T	Solar radiation on the PV array [kW/m ²]
$G_{T,STC}$	Incident Radiation at STC [1 kW/m ²]
U_{hub}	Wind speed at the hub height of the wind turbine [m/s]
U_n	Measured wind speed at anemometer height [m/s]
Z_{hub}	Hub height of the wind turbine [m]
Z_{anem}	The anemometer height [m]
Z_0	The surface roughness length [m]
ρ	The actual air density [kg/m ³]
ρ_0	The air density at STP [1.225 kg/m ³]
P_{WTG}	Power output from the power curve at STP [kW]
P_n	The wind turbine power output [kW]
NPC	Net Present cost
T_{AC}	Total Annualized cost
E_{served}	Total electrical load served [kWh/yr]
COE	Cost of Energy/Electricity
CRF	Capital Recovery Factor.

MODEL LOCATION

Lagos, one of the largest countries in Africa and located in Nigeria, was used for the model. Lagos has coordinates 6.4531°N and 3.3958° E with an estimated population of 17.5 million as at 2006 census [6]. The electricity infrastructure in Lagos, as the main commercial city in Nigeria is inadequate and this has resulted in power outages, blackouts, and high technical and economic losses on a daily basis [7]. From the inception of electricity production in Nigeria, electric power stations depend on water, gas, coal as fuel sources. Current reforms put in place by Nigerian government have allowed privately generated electricity and encourage the use of renewable energy.

HES COMPONENTS AND RESOURCES

Load Demand

A hypothetical load for the model location used is shown in Fig 1. The load demand is assumed to be constant throughout the year but can be varied from one location to another. This is assumed for a typical Lagos neighborhood of 20 to 30 households with average demand of 8.4 kWh per day.

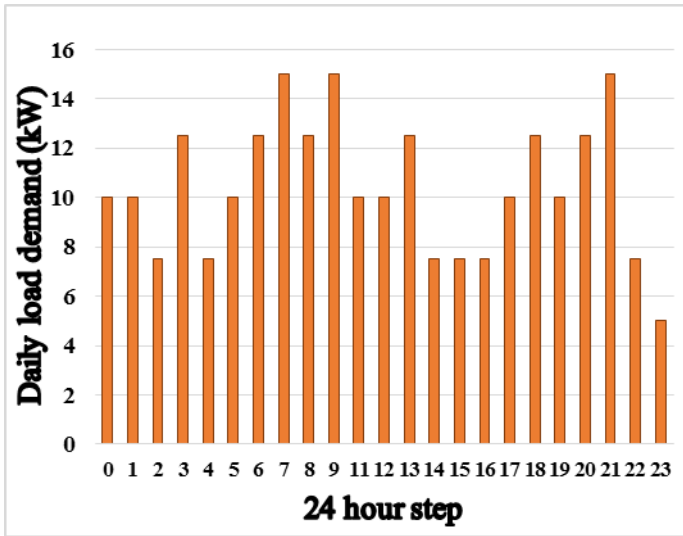


Fig 1: Average Load demand for a typical day

Unreliable Grid

Unreliable grid is modeled by setting minimum load capacity of the generator to 0%, all costs set to zero and the grid size is made large. Grid power has 1kg of fuel equals to 1kWh, the price per kg now equals price per kWh. In other to function as a grid, the energy content of the fuel is set to 3.6MJ/kg, density to 1000kg/m³ and reports generator consumption in kg. The generator is made 100% fuel efficient machine by setting intercept coefficient to 0 and the slope to 1 [8]. Fig 2 shows generator scheduling capacity that allows the Grid to be forced on/off.

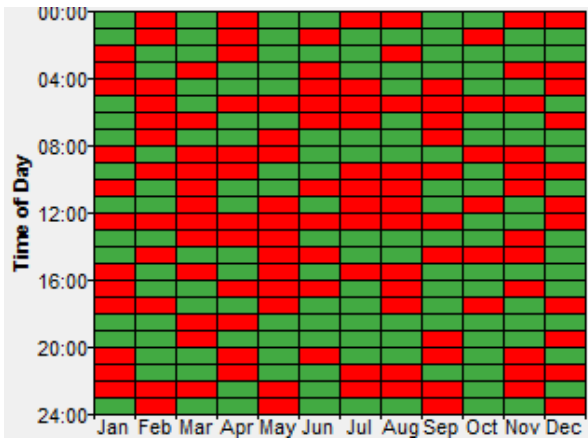


Fig 2: Forcing off of Generator to simulate unreliable Grid in HOMER.

Red and green in Fig 2 depict time of the day when there is electricity outage and electricity availability respectively. This modification to generator component makes the simulation of frequent outages possible and also act as unreliable grid for this research.

Solar PV components

Solar irradiance for Lagos coordinates 6.4531°N and 3.3958°E was retrieved from HOMER which corresponds to the coordinates as shown in Fig 3. Annual average global radiation is 4.75066 kW/m²/day. Solar PV inputs are derating factor of 80%, ground reflectance of 20% and PV lifetime of 20 years. Power output for Solar PV is expressed below without considering the effect of temperature:

$$P_{PV} = Y_{PV} f_{PV} \left(\frac{G_T}{G_{T,STC}} \right) \quad (1)$$

Wind Turbine Components

Average monthly wind speed (10m anemometer height) as analyzed by Ahmed et al [9] was used as shown in Fig 3. A generic 10 kW turbine was used for the model. Fig 4 shows the power curve of the turbine with cut-in speed of 3 m/s (power can only be produced with wind speed above 3 m/s) and cut-out wind speed of 24 m/s (no power can be drawn from the turbine at wind speed greater than 24 m/s). Wind Turbine inputs are Weibull parameter of 2, 1-hr autocorrelation factor of 0.85, diurnal pattern strength of 0.25 and Hour of peak wind speed of 15. Hub height of the turbine is 25m with lifetime of 25 years. HOMER used a time step of 60 minutes to generate 8760 wind speed. The following shows the expression that HOMER uses to generate power output of the turbine:

$$U_{hub} = U_n \cdot \frac{\ln(Z_{hub}/z_o)}{\ln(Z_{anem}/z_o)} \quad (2)$$

Corresponding power output for wind speed at each time is obtained from the power curve shown in Fig 4.

$$P_n = \left(\frac{\rho}{\rho_o} \right) P_{WTG} \text{ For } n=1, 2, 3, \dots, 8760 \text{ hrs.} \quad (3)$$

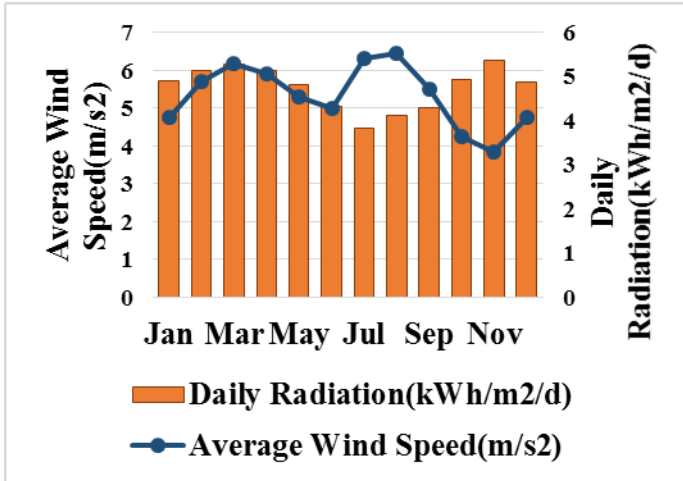


Fig 3: Daily Radiation and Monthly Wind Speed

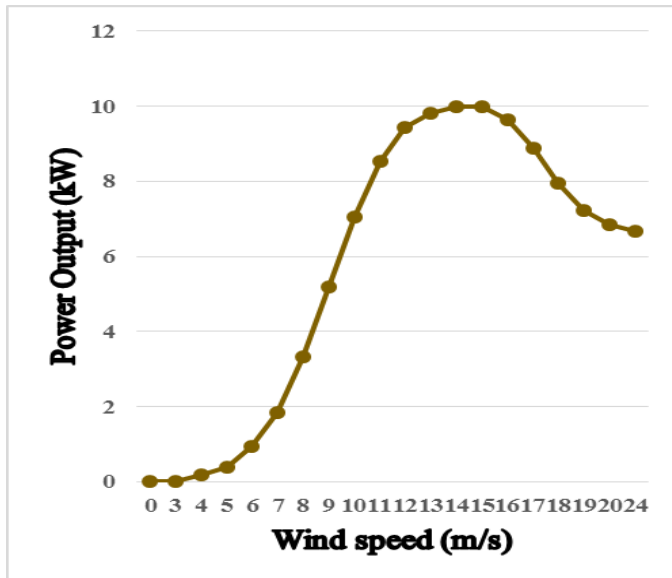


Fig 4: Power curve of 10 kW Generic Turbine

Diesel generator

A diesel generator is employed in this model to serve as back up. This comes in to play to reflect real life situation in most developing countries where electricity outage could occur for more than 24 hours. At this time, grid and renewable energy sources may fail to meet electricity demand at longer outage and Diesel generator will kick in to supply electricity.

Decision criteria for the system

Net Present Cost (NPC)

This is the life cycle cost of the HES. NPC factors is all the cost occurring within the project life time – Initial cost, Replacement cost, operation and maintenance cost and fuel cost for the system. HOMER calculates NPC with the following equation [10]

$$NPC(\$) = \frac{T_{AC}}{CRF} \quad (4)$$

Where T_{AC} is the total annualized cost which contains Initial cost, Replacement cost, operation and maintenance cost and fuel cost for the system and Capital Recovery factor (CRF) is given by [10]

$$CRF = \frac{i(1+i)^N}{(1+i)^N - 1} \quad (5)$$

Where N is the project life time and i is the annual real interest rate ($i = 12\%$ for this paper based on current inflation rate of the model location).

Cost of Energy (COE)

HOMER calculates the cost of energy (COE) as the average cost per kWh of useful electrical energy produced by the system and is given below mathematically [10]

$$COE\left(\frac{\$}{kWhr}\right) = \frac{T_{AC}}{E_{served}} \quad (6)$$

Excess electricity

Excess electricity is the surplus electricity generated by the system during peak time. Excess electricity occur after load demand has been met. Excess Electricity (EE) is expressed mathematically below,

$$EE = Total\ Electricity\ generated - E_{served} \quad (7)$$

MODELS CONSIDERED

When designing an HES, different scenarios and conditions need to be considered to factor in all possibilities. Three HES model scenarios were considered in this study:

- Model 1 – This consists of Intermittent Grid and Back up Diesel Generator. This is the

existing and common set up in model location. Most electricity consumers in Lagos, Nigeria depend on generator to supplement inefficient electricity supply from the Grid

- ii. Model 2 – This consists of intermittent Grid, Solar PV and Back up Diesel Generator.
- iii. Model 3 – This consists of intermittent Grid, Wind Turbine, and Solar PV and Back up Diesel Generator.

HOMER RESULTS AND DISCUSSION

Simulation of different models was carried out with a project life time of 25 years. A backup generator is employed in all the model scenario to maintain system balance and to take care of worst case scenario when intermittent grid and renewable energy sources are down. HOMER software helped to analyze the best possible combination from economic perspective as shown in the cost summary in Table 1. It is important to note that stand alone HES was not modeled because cost of electricity for Grid in current model location, Lagos, is very cheap (\$0.18) compared to other cities around the world. Based on this, many consumers still connect to grid and utilize it when it is available.

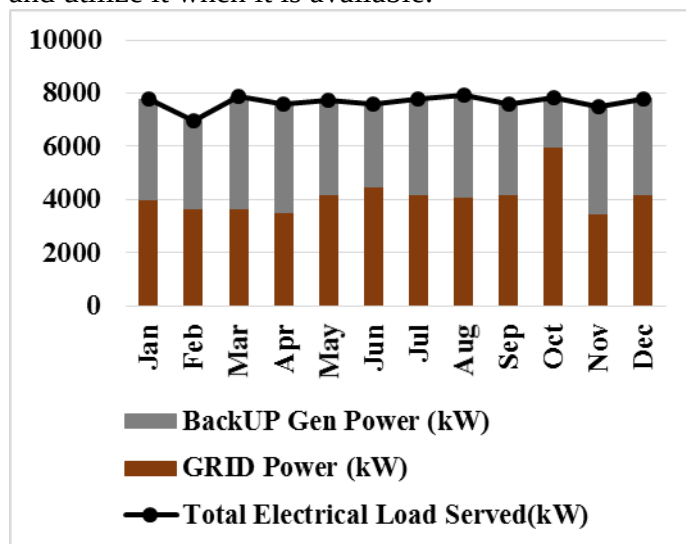


Fig 5: Monthly Load demand satisfaction by Generator and intermittent Grid (Model 1)

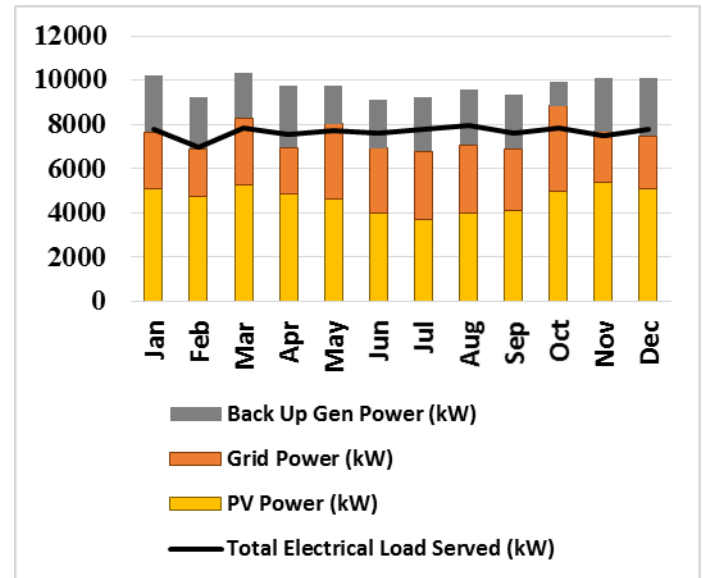


Fig 6: Monthly Load demand satisfaction by Generator, Solar PV and Intermittent Grid (Model 2).

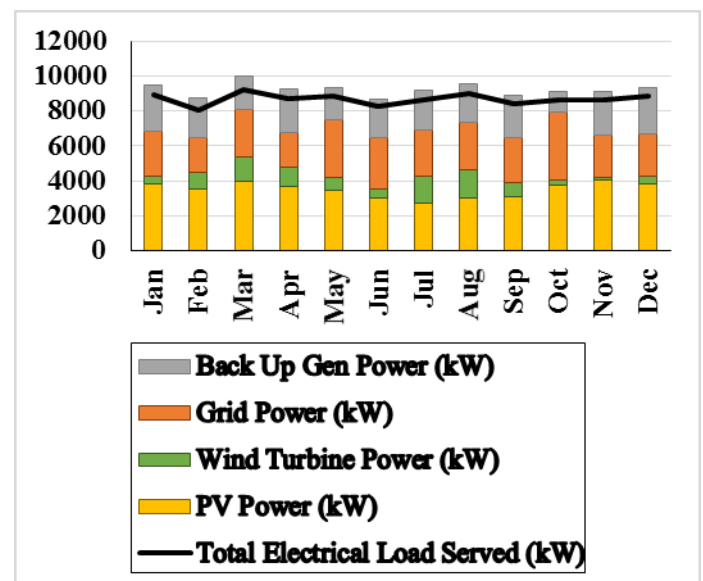


Fig 7: Monthly Load demand satisfaction by Solar PV, Wind Turbine, Generator and intermittent Grid (Model 3)

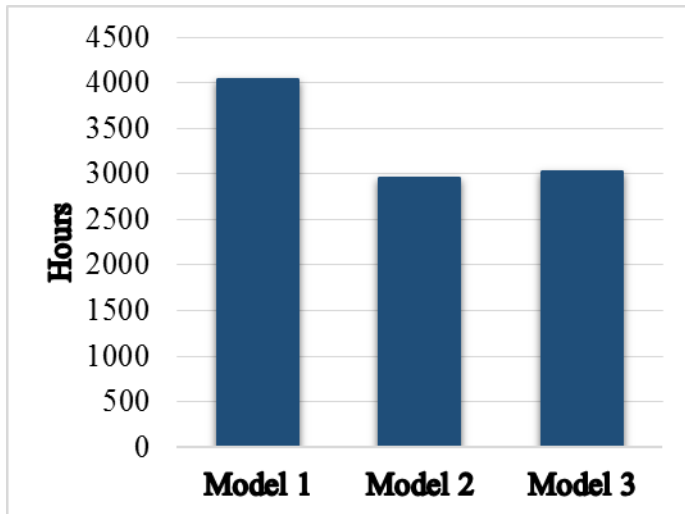


Fig 8: Generator operating hour for each model for a typical year.

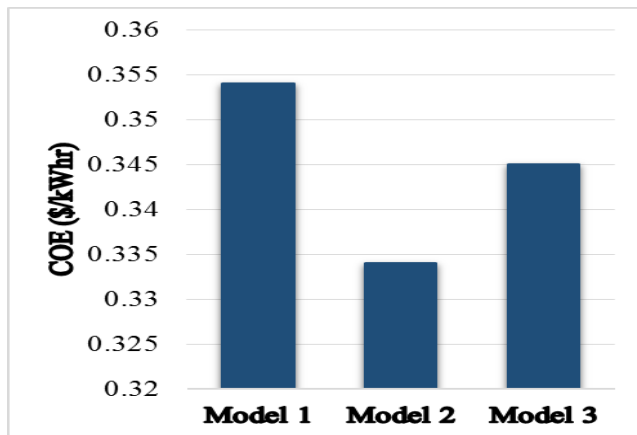


Fig 9: Cost of Energy for different model

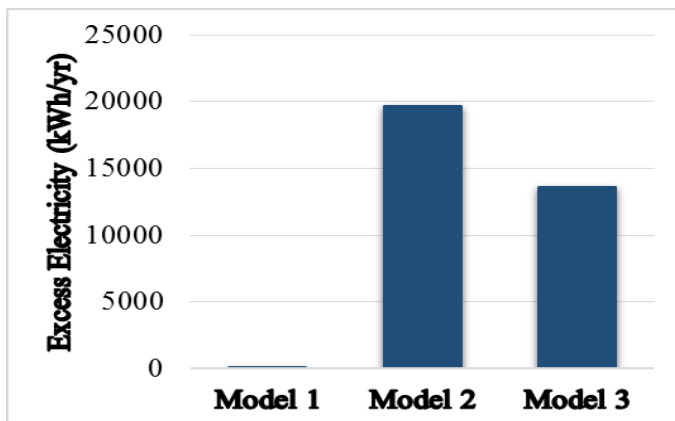


Fig 10: Excess electricity generated by each model

Table 1: Breakdown of Cost Summary for each Model Scenario

Model scenario			
Cost Summary	Model 1	Model 2	Model 3
Initial Capital Cost(\$)	6800	97100	98200
Operating Cost(\$)	31561	21304	22256
Annualized cost (\$)	23616	35520	36634
Total NPC(\$)	222363	242604	250210
COE (\$)	0.354	0.334	0.345

Table 2: Percentage Electrical Production for each components in Model 1

Model 1		
Electricity Production	kWh/yr	%
Grid	49,671	54
BackUP	42,433	46
Total	92,104	100

Table 3: Percentage Electrical Production for each components in Model 2

Model 2		
Electricity Production	kWh/yr	%
PV array	55,932	48
Grid	33,621	29
BackUP	27,151	23
Total	116,704	100

Table 4: Percentage Electrical Production for each components in Model 3

Model 3		
Electricity Production	kWh/yr	%
PV array	41,949	38
Wind turbine	10,038	9
Grid	32,166	29
BackUP	27,546	25
Total	111,698	100

As shown in Figure 5, load demand was satisfied for each month of the year and Table 2 shows the

percentage contribution of intermittent grid and diesel generator. Fig 6 shows load demand satisfaction for the combination of solar PV and back-up generator connected to intermittent grid. Percentage electricity production of each components is shown Table 3. Monthly Load demand satisfaction for Model 3 is shown in Fig 7 with percentage electricity production contribution from each components as shown in Table 4. All The system was simulated to meet maximum capacity. Models 2 and 3 exceeded monthly load demand to produce excess electricity. Daily load demand was assumed to be constant throughout the year because there is no significant change in rate of consumption for different time of the year.

It can be seen that Solar PV supplied bulk of the electricity production in the HES connected with intermittent grid as shown in Tables 3 and 4. For the model 3 with wind turbine, much electricity was not produced because the cut-in wind speed of the turbine is 3 m/s (i.e No electricity production can be generated from the wind turbine when the average wind speed is less than 3 m/s).

Overall results of all the model scenario indicates that combination Solar PV, Backup generator connected with intermittent grid(Model 2) is the optimal system for Lagos based on COE, excess electricity generated by the system and operating hours of generator as shown in Fig 8,9 and 10. It should be noted that reduced operating hours will translate to reduction in carbon emission and noise pollution from the generator which is in line with the aim of this research.

Model 1 is currently used in the model location to meet the shortfall from grid. It is shown that HES connected with intermittent grid can effectively and efficiently meet the shortfall in load demand and still offer most developing countries the opportunity to participate in global transition to clean and reliable energy generating system.

CONCLUSION

HES has become an applicable solution to the challenges that the world faces today for sustainable energy supply and environmental protection. HES

can be developed and optimized in order to suit the needs of any location based on type of energy requirements, available resources and topography of the location. The optimal sizing of HES can significantly improve the economic and technical performance of power supply as well as promoting the widespread use in developing countries.

In this research (which is still ongoing), HES connected with intermittent grid was optimized using HOMER software based on an hourly daily load demand for our model location. The simulation models were carried out based on the mean solar radiation and wind speed data for Lagos, Nigeria. HOMER software is used to optimize the sizes of the system design parameters. The optimization is done with three different modeling scenarios.

The methodology described in this study provides an important and systematic approach for design and analysis of HES connected with intermittent grid. The simulation models of this study can be replicated for any other location and extended in many directions by considering energy storage and sellback of excess electricity to the grid. This research is in the process of being extended as a future research with regard to those directions. Also, experimental analysis of all these models are in process to validate the results and show error analysis of HOMER software.

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